

IMAN BAREKATAIN

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Iman Barekatain

Education

SUMA double degree in sustainable materials:

- KU Leuven (#1 university in Belgium and [#48](#) globally)

M.Sc. in Materials Engineering

- University of Milan-Bicocca (#10 university in Italy and [#299](#) globally)

M.Sc. in Materials Science and Nanotechnology

September 2023 – Now (last semester)

Bachelor's degree in Materials Engineering

Isfahan University of Technology ([#4](#) university in Iran, [#338](#) globally for Materials science)

B.Sc. in Metallurgy and Materials Engineering

September 2018 - February 2023



Research Experience

Master's thesis

Effect of laser beam shaping on Mn, N-stabilized stainless steels manufactured using L-PBF

An investigation of the effect of laser distribution profiles, particularly Gaussian vs. ring, on microstructure and vaporization behavior of austenitic stainless steels stabilized with manganese and nitrogen, via a joint simulation and experiment approach.

•Promoter: Prof. Kim Vanmeensel (KU Leuven)

Bachelor's thesis

Artificial intelligence in materials science and engineering

A report of recent advances and applications of machine learning in solid state materials science.

•Advisor: Prof. Mahmood Meratian (IUT)

Article

An experiment on enhancing properties of a ceramic matrix composite using certain nano additives. (postponed)

- Learned and employed **SPS method** and related parameters.
- Performed Archimedes density test, pin-on-disk wear test, compression test, Charpy impact test.
- Learned about different methods of powder metallurgy and powder sintering as well as chemical dispersion methods.

Research Interests

- Composites
- Artificial Intelligence
- Powder metallurgy
- Sustainability
- Additive manufacturing
- Biomaterials

Language Skills

English: Advanced

•**IELTS academic band score: 8** — Reading: 9, Listening: 8.5, Speaking: 7, Writing: 7

Persian: Native

Work Experience

Internship in materials development at [Toyota Motor Europe](#) (TME)

2025

A 3-month internship in Toyota's main R&D hub in Europe – located in Zaventem, Belgium – in developing a new 3D printed material. The details are subject to NDA.



Web developer and UI designer at [Libratech IT Solutions](#), Esfahan, Iran

2018 – 2022

- Part-time participation, including making websites and web-based applications using HTML, CSS, JavaScript, VueJs platform and WordPress; as well as user interface design using Adobe XD.



Internship in R&D section at [Tamkar Industrial Group](#), Esfahan, Iran

2022

- Including research, supervision, translation and communication with foreign partners.



Skills

Materials characterization techniques: XRD, XRF, TEM, SEM, OM, AFM.

Mechanical properties tests: Hardness, Toughness, Wear, Tensile.

Academic software: imageJ, Key to steel, COMSOL Multiphysics, Thermo-Calc.

Programming languages: JavaScript (VueJS, basic), Python (basic), HTML, CSS.

Computer skills: Microsoft office, Adobe photoshop, Adobe Illustrator, Adobe XD.

Other: WordPress, software development using AI.

Voluntary Experience

Member of *Saleh NGO charity*

2018–2023

Charity for providing facilities for orphaned children and holding celebrations for them.

Active member of *Aria Cultural Center*

2018-2020

Holding celebrations, tours and environmental games in Isfahan University of Technology.

- Contributed to organization, preparation and arrangement of Yaldā Night celebration in 2018 with 1000+ guests.
- Contributed to building a mobile scene on a tractor-trailer for music performance.
- Contributed to arrangement of Karino breakfast contest in 2018 and 2019.

Member of *Theater Cultural Center*

2018–2020

- Contribution in selling tickets, arrangement and conduction of ‘*All thieves are not thieves*’ theater.
- Council member of theater cultural center in Isfahan university of technology in 2019.

Honors and Awards

Top 1% rank in national university entrance exam (B.Sc. konkur) in mathematical physics

2018

Top 5% GPA in the last 4 semesters of B.Sc. education in Isfahan University of Technology materials department (average 17/20)

2020 – 2022

Honors and Awards (continued)

Distinguished B.Sc. student in the Department of Materials Engineering at Isfahan University of Technology

2020 – 2022

Intern of the month in Tamkar Industrial Co.

2022

Web blogging contest winner in high school

2014

References

Prof. Kim Vanmeensel ([Scholar profile](#), [Website](#))

Associate professor at Faculty of Engineering Science, KU Leuven.

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- Phone: +32 16 32 13 14

Prof. Mahmood Meratian ([Scholar profile](#), [Website](#))

Associate professor at Materials Science and Engineering Department, Isfahan University of Technology

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Aurelie Serre

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